Mysql task

Using MySQL, design a database whose name is IMDB. Create proper MySQL tables, Primary Key, Foreign Key, add data into the MySQL tables and do the

```
create database IMDB;
use IMDB;

-- Movies
create table Movies (
   movie_id INT PRIMARY KEY AUTO_INCREMENT,
   title VARCHAR(200),
   release_year INT
);

insert into Movies (title, release_year) values
('Baahubali: The Beginning', 2015),
('Enthiran', 2010),
('Vikram', 2022),
('Kaithi', 2019),
('Ponniyin Selvan: Part 1', 2022);
```

movie_id	title	release_year
1	Baahubali: The Beginning	2015
2	Enthiran	2010
3	Vikram	2022
4	Kaithi	2019
5	Ponniyin Selvan: Part 1	2022
NULL	NULL	NULL

```
-- Media (Video/Image linked to Movie)
create table Media (
```

```
    media_id INT PRIMARY KEY AUTO_INCREMENT,
    movie_id INT,
    media_type ENUM('Video','Image'),
    url VARCHAR(255),
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id)
);

insert into Media (movie_id, media_type, url) values
(1, 'Video', 'http://trailers.com/baahubali.mp4'),
(1, 'Image', 'http://images.com/baahubali.jpg'),
(2, 'Video', 'http://trailers.com/enthiran.mp4'),
(2, 'Image', 'http://images.com/enthiran.jpg'),
(3, 'Video', 'http://trailers.com/vikram.mp4'),
(3, 'Image', 'http://images.com/vikram.jpg'),
(4, 'Video', 'http://trailers.com/kaithi.mp4'),
(5, 'Video', 'http://trailers.com/ps1.mp4'),
(5, 'Image', 'http://images.com/ps1.jpg');
```

media_id	movie_id	media_type	url
1	1	Video	http://trailers.com/baahubali.mp4
2	1	Image	http://images.com/baahubali.jpg
3	2	Video	http://trailers.com/enthiran.mp4
4	2	Image	http://images.com/enthiran.jpg
5	3	Video	http://trailers.com/vikram.mp4
6	3	Image	http://images.com/vikram.jpg
7	4	Video	http://trailers.com/kaithi.mp4
8	5	Video	http://trailers.com/ps1.mp4
9	5	Image	http://images.com/ps1.jpg
NULL	NULL	NULL	NULL

```
-- Genres
create table Genres (
    genre_id INT PRIMARY KEY AUTO_INCREMENT,
    genre_name VARCHAR(100)
```

```
);
insert into Genres (genre_name) values
('Action'),
('Drama'),
('Thriller'),
('Historical'),
('Science Fiction');
```

genre_id	genre_name
1	Action
2	Drama
3	Thriller
4	Historical
5	Science Fiction
NULL	NULL

Movie-Genre relationship (many-to-many)

```
create table MovieGenres (
   movie_id INT,
   genre_id INT,
   PRIMARY KEY(movie_id, genre_id),
   FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
   FOREIGN KEY (genre_id) REFERENCES Genres(genre_id)
);

insert into MovieGenres (movie_id, genre_id) values
(1, 1), (1, 2),
(2, 5),
(3, 1), (3, 3),
(4, 1), (4, 3),
(5, 2), (5, 4);
```

movie_id	genre_id
1	1
3	1
4	1
1	2
5	2
3	3
4	3
5	4
2	5
NULL	NULL

```
-- Users
create table Users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100)
);

insert into Users (name) values
('Arun'),
('Priya'),
('Karthik'),
('Divya');
```

user_id	name
1	Arun
2	Priya
3	Karthik
4	Divya
NULL	NULL

```
-- Reviews
create table Reviews (
    review_id INT PRIMARY KEY AUTO_INCREMENT,
    movie_id INT,
```

```
    user_id INT,
    review_text TEXT,
    rating INT,
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

insert into Reviews (movie_id, user_id, review_text, rating) values
(1, 1, 'Epic visuals and storytelling!', 5),
(2, 2, 'Rajinikanth was amazing!', 4),
(3, 3, 'Intense and gripping thriller.', 5),
(4, 4, 'Lokesh Kanagaraj delivers again.', 4),
(5, 1, 'Beautiful adaptation of Kalki novel.', 5);
```

review_id	movie_id	user_id	review_text	rating
1	1	1	Epic visuals and storytelling!	5
2	2	2	Rajinikanth was amazing!	4
3	3	3	Intense and gripping thriller.	5
4	4	4	Lokesh Kanagaraj delivers again.	4
5	5	1	Beautiful adaptation of Kalki novel.	5
NULL	NULL	NULL	NULL	NULL

```
-- Artists
create table Artists (
    artist_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100)
);

insert into Artists (name) values
('Rajinikanth'),
('Kamal Haasan'),
('Suriya'),
('Karthi'),
```

('Aishwarya Rai Bachchan');

artist_id	name
1	Rajinikanth
2	Kamal Haasan
3	Suriya
4	Karthi
5	Aishwarya Rai Bachchan
NULL	NULL

```
-- Skills
create table Skills (
    skill_id INT PRIMARY KEY AUTO_INCREMENT,
    skill_name VARCHAR(100)
);

insert into Skills (skill_name) values
('Acting'),
('Dancing'),
('Singing'),
('Stunts');
```

skill_id	skill_name
1	Acting
2	Dancing
3	Singing
4	Stunts
NULL	NULL

```
-- Artist-Skill relationship
create table ArtistSkills (
    artist_id INT,
    skill_id INT,
    PRIMARY KEY(artist_id, skill_id),
    FOREIGN KEY (artist_id) REFERENCES Artists(artist_id),
```

```
    FOREIGN KEY (skill_id) REFERENCES Skills(skill_id)
);

insert into ArtistSkills (artist_id, skill_id) values
(1, 1), (1, 2),
(2, 1), (2, 3),
(3, 1), (3, 4),
(4, 1), (4, 4),
(5, 1), (5, 2);
```

artist_id	skill_id
1	1
2	1
3	1
4	1
5	1
1	2
5	2
2	3
3	4
4	4
NULL	NULL

```
-- Roles in Movies
create table Roles (
    role_id INT PRIMARY KEY AUTO_INCREMENT,
    role_name VARCHAR(100)
);

insert into Roles (role_name) values
('Hero'),
('Villain'),
('Supporting Actor'),
('Director');
```

role_id	role_name
1	Hero
2	Villain
3	Supporting Actor
4	Director
NULL	NULL

```
-- Artist performing roles in Movies
create table MovieRoles (
    movie_id INT,
    artist_id INT,
    role_id INT,
    PRIMARY KEY(movie_id, artist_id, role_id),
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
    FOREIGN KEY (artist_id) REFERENCES Artists(artist_id),
    FOREIGN KEY (role_id) REFERENCES Roles(role_id)
);

insert into MovieRoles (movie_id, artist_id, role_id) values
(2, 1, 1),
(3, 2, 1),
(4, 4, 1),
(5, 5, 1),
(1, 3, 2);
```

movie_id	artist_id	role_id
2	1	1
3	2	1
1	3	2
4	4	1
5	5	1
NULL	NULL	NULL

-- **1. movie should have multiple media**

```
select Movies.title AS movie,
    Media.media_type,
    Media.url as media_url
from Movies
join Media
on Movies.movie_id = Media.movie_id
order by Movies.title, Media.media_type;
```

movie	media_type	media_url
Baahubali: The Beginning	Video	http://trailers.com/baahubali.mp4
Baahubali: The Beginning	Image	http://images.com/baahubali.jpg
Enthiran	Video	http://trailers.com/enthiran.mp4
Enthiran	Image	http://images.com/enthiran.jpg
Kaithi	Video	http://trailers.com/kaithi.mp4
Ponniyin Selvan: Part 1	Video	http://trailers.com/ps1.mp4
Ponniyin Selvan: Part 1	Image	http://images.com/ps1.jpg
Vikram	Video	http://trailers.com/vikram.mp4
Vikram	Image	http://images.com/vikram.jpg

-- **2.Movie can belong to multiple genre**

```
select Movies.title AS movie,
    Genres.genre_name
from Movies
join MovieGenres
on Movies.movie_id = MovieGenres.movie_id
join Genres
on MovieGenres.genre_id = Genres.genre_id
order by Movies.title, Genres.genre_name;
```

movie	genre_name
Baahubali: The Beginning	Action
Baahubali: The Beginning	Drama
Enthiran	Science Fiction
Kaithi	Action
Kaithi	Thriller
Ponniyin Selvan: Part 1	Drama
Ponniyin Selvan: Part 1	Historical
Vikram	Action
Vikram	Thriller

-- 3. Movie can have multiple review and review can belong to user

```
select Movies.title, Users.name AS reviewer, Reviews.review_text, Reviews.rating
from Movies
join Reviews
on Movies.movie_id = Reviews.movie_id
join Users
on Reviews.user_id = Users.user_id
order by Movies.title, Users.name;
```

title	reviewer	review_text	rating
Baahubali: The Beginning	Arun	Epic visuals and storytelling!	5
Enthiran	Priya	Rajinikanth was amazing!	4
Kaithi	Divya	Lokesh Kanagaraj delivers again.	4
Ponniyin Selvan: Part 1	Arun	Beautiful adaptation of Kalki novel.	5
Vikram	Karthik	Intense and gripping thriller.	5

-- **4. Artist can have multiple skills**

```
select Artists.name AS artist, Skills.skill_name
from Artists
join ArtistSkills
on Artists.artist_id = ArtistSkills.artist_id
join Skills
on ArtistSkills.skill_id = Skills.skill_id
```

order by Artists.name, Skills.skill_name;

artist	skill_name
Aishwarya Rai Bachchan	Acting
Aishwarya Rai Bachchan	Dancing
Kamal Haasan	Acting
Kamal Haasan	Singing
Karthi	Acting
Karthi	Stunts
Rajinikanth	Acting
Rajinikanth	Dancing
Suriya	Acting
Suriya	Stunts

```
select Artists.name AS artist, count(Skills.skill_id) AS total_skills
from Artists
join ArtistSkills
on Artists.artist_id = ArtistSkills.artist_id
join Skills
on ArtistSkills.skill_id = Skills.skill_id
group by Artists.name;
```

artist	total_skills
Rajinikanth	2
Kamal Haasan	2
Suriya	2
Karthi	2
Aishwarya Rai Bachchan	2

-- **5. Artist can perform multiple role in a single film**

```
select Movies.title AS movie,
    Artists.name AS artist,
    Roles.role_name
from Movies
```

```
join MovieRoles
on Movies.movie_id = MovieRoles.movie_id
joib Artists
on MovieRoles.artist_id = Artists.artist_id
join Roles
on MovieRoles.role_id = Roles.role_id
order by Movies.title, Artists.name, Roles.role_name;
```

movie	artist	role_name
Baahubali: The Beginning	Suriya	Villain
Enthiran	Rajinikanth	Hero
Kaithi	Karthi	Hero
Ponniyin Selvan: Part 1	Aishwarya Rai Bachchan	Hero
Vikram	Kamal Haasan	Hero

```
select Movies.title AS movie,
      Artists.name AS artist,
      (select group_concat(Roles.role_name SEPARATOR ', ')
       from MovieRoles
       join Roles
       on MovieRoles.role_id = Roles.role_id
       where MovieRoles.movie_id = Movies.movie_id
         and MovieRoles.artist_id = Artists.artist_id) AS roles
from Movies
join MovieRoles
on Movies.movie_id = MovieRoles.movie_id
join Artists
on MovieRoles.artist_id = Artists.artist_id
order by Movies.title, Artists.name;
```

movie	artist	roles
Baahubali: The Beginning	Suriya	Villain
Enthiran	Rajinikanth	Hero
Kaithi	Karthi	Hero
Ponniyin Selvan: Part 1	Aishwarya Rai Bachchan	Hero
Vikram	Kamal Haasan	Hero